

RATIONAL NUMBERS AND ABSOLUTE VALUE



LESSON INTRODUCTION

6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers.

6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite directions using rational numbers. Compare them on a number line and explain the meaning of zero within its context.

6.NSO.1.3 Given a mathematical or real-world context, interpret the absolute value of a number as the distance from zero on a number line. Find the absolute value of rational numbers.

LEARNING TARGETS

- I can plot, compare, and order rational numbers in a mathematical or real-world context.
- I can compare and perform operations involving the absolute value of two numbers.

BENCHMARK COHERENCE

BUILDING ON

MA.5.NSO.1.4 Plot, order, and compare multi-digit numbers with decimals up to the thousandths place.

WORKING TOWARD

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

ESSENTIAL QUESTIONS

- How can rational numbers be compared and ordered when expressed as both fractions and decimals?
- How are absolute values simplified?
- How can absolute values be compared?

LESSON NARRATIVE

Students review rational numbers and absolute value in preparation for tomorrow's work with operations and integers. They simplify and model values using number lines and utilize symbols for comparing. Students summarize the concepts throughout, helping to consolidate the information that is presented in this lesson.

COMPONENT SUMMARY

1.3.1 WARM-UP (~5 MIN.)

Solve and justify a conversion problem

1.3.3 GUIDED PRACTICE (10 MIN.)

Review how to plot rational numbers on a number line

1.3.5 TRY IT (10 MIN.)

Practice finding, modeling, and comparing absolute values

1.3.2 REFRESHER (5 MIN.)

Review definition of a rational number

1.3.4 GUIDED PRACTICE (10 MIN.)

Explore and compare absolute values

1.3.6 WRAP-UP (~5 MIN.)

Self-assess learning by identifying concepts understood and questions that remain

RESOURCES

GLOSSARY

Key Terms:

- Rational numbers

Additional Terms:

- Integers
- Absolute value

MATERIALS

- (Optional) Enlarged number lines

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students might not recognize that the absolute value symbol is a grouping symbol. This could cause confusion when students are introduced to it in 1.3.4 Guided Practice and when simplifying and comparing absolute values during the remainder of the lesson.

THINGS TO CONSIDER

- Students in grade 7 are expected to perform all four operations with rational numbers with procedural fluency. Balancing this with each student's gap in learning and understanding may be difficult. Consider establishing a chart to keep account of fluency milestones. Keep in mind fluency should not be based on speed.
- This lesson is a review of concepts learned in grades 5 and 6, meant to prepare students to review operations using integers in the next lesson. Consider leveraging On-Ramp to 6th Grade video or practice tools to provide additional remediation opportunities for students who struggle to compare values of rational numbers using inequalities.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

Students are presented with an opportunity in 1.3.3 Guided Practice question 1c to connect their work with fractions, comparing and ordering rational numbers, and the number line when asked to plot $-1\frac{7}{12}$. Consider utilizing a Think Aloud to demonstrate how students might approach the task and use what they already know that might help them. Pose questions and encourage students to reflect and respond.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Clarifications in MA.K12.MTR.7.1 note that mathematicians who use patterns and structure "...relate previously learned concepts to new concepts." Utilize the discussion prompts throughout to facilitate student understanding of these connections and how they apply to the concepts being explored in the lesson.

DIFFERENTIATE INSTRUCTION

Life-Size Number Line

Does your classroom have tiles or gridlines on the floor that could be used as life-size number lines? If so, consider transforming your classroom into Number Line Land, using the gridlines as kinesthetic playgrounds for students to make sense of number lines as they plot and order rational numbers. Alternatively, utilize other methods like enlarged number lines, chart paper, or similar tools for students to have a variety of entry points to this lesson.

1.3.1 WARM-UP: MATH TALK

TEACHER MOVES

Consider providing conversion charts for students to make use of tools during the Warm-Up.



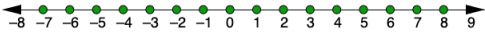
1.3.1 Warm-Up: Math Talk

1. Are there more inches in a mile or seconds in a day?
There are more seconds in a day since there are 86,400 seconds in one day and 63,360 inches in one mile.
2. Discuss with your partner and justify your reasoning mathematically.
Answers will vary. There are only 63,360 inches in a mile, and 86,400 seconds in a day.



1.3.2 Refresher: Rational Numbers, Integers, and Negative Numbers

The values in green on the number line are some of the values in the set of **integers**.



The set of **rational numbers** includes all integers as well as the fractional parts between integers on the number line.



Rational numbers are numbers that can be expressed as the ratio of two integers.

1. Use the definition of rational numbers to write each number as a ratio of two integers.

Number	Ratio of Two Integers
-1.5	$-\frac{3}{2}$
9	$\frac{9}{1}$

1.3.2 REFRESHER: RATIONAL NUMBERS, INTEGERS, AND NEGATIVE NUMBERS

TEACHER MOVES

Students were introduced to integers and operations with integers in grade 6. In grade 6, students performed operations on positive fractions and decimals but did not perform operations on negative fractions and decimals. Because fluency of operations with rational numbers is one of the five areas that the BEST standards emphasize in grade 7 ensuring students have the foundational understanding of rational numbers is the purpose of this lesson. Students will encounter rational numbers throughout the course.

ENRICHMENT

- Consider prompting questions to activate prior knowledge while positioning students for success in ordering these numbers in the next activity.
- What do you know about ratios? What do they look like?
 - Are fractions ratios? Why or why not?
 - How might you represent a number like -1.5 or 9 in terms of a ratio?

1.3.3 GUIDED PRACTICE: PLOTTING RATIONAL NUMBERS ON THE NUMBER LINE

DIFFERENTIATE INSTRUCTION

Consider enlarging the number line or constructing a life-size number line using any tiles or gridlines on the classroom floor to create kinesthetic entry points for students.

TEACHER MOVES

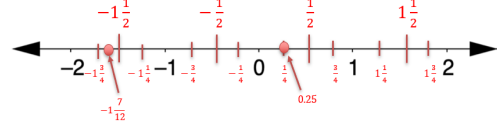
Compare and Connect

Consider your students when planning this activity. A teacher-led or student-led Think Aloud may provide mental modeling for students to explore different entry points.

- What is a “benchmark” fraction that I am familiar with that would represent halfway between each value?
- Is there a benchmark fraction that is close to ___?
- How does ___ compare to ___?
- Is ___ closer to ___ or to the next whole number?
- How can this information help to determine where to plot ___?

1.3.3 Guided Practice: Plotting Rational Numbers on the Number Line

1. Use the number line shown to complete the following.

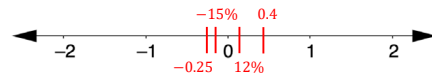


- Make a tick mark halfway between each value marked on the number line. Label each tick mark with the value of its location.
- Make another tick mark halfway between each value and the tick marks that were previously made. Label each with the value of its location.
- Plot and label the values $-1\frac{7}{12}$ and 0.25 on the number line.

2. Four values are shown.

0.4 -15% 12% -0.25

- Plot each of the values on the number line. Label each point with its numeric value.



- Complete the inequality statement to show the relationship between the values.

-0.25 < -15% < 12% < 0.4

3. Complete the statements.

a. When moving from left to right ($\rightarrow$) on the number

line, the values

- ☐ increase.
☐ decrease.

This can be

described as moving in a

- ☐ positive
☐ negative

direction.

b. When moving from right to left ($\leftarrow$) on the number

line, the values

- ☐ increase.
☐ decrease.

This can be

described as moving in a

- ☐ positive
☐ negative

direction.

4. Order these values from least to greatest.

0.43, -4.2, 0, 50%, -1.5, -6.712, 19%

-6.712, -4.2, -1.5, 0, 19%, 0.43, 50%

1.3.3 GUIDED PRACTICE: PLOTTING RATIONAL NUMBERS ON THE NUMBER LINE (CONTINUED)

ENRICHMENT

Prompt students to think about how to best compare the values given in question 2.

- How can decimals be rewritten as percentages?
- How can percentages be rewritten as decimals?
- Why might it be helpful if all values were written as either decimals or percentages?
- Is it better to compare if all are written as decimals or as percentages? Why?

TEACHER MOVES

Students may respond to question 4 using all decimals or percentages, depending on which representation they used to compare. This is an opportunity to discuss equivalent values and to value all answers that are equivalent. Understanding equivalency of numeric values empowers students to have flexible thinking. Consider incorporating different equivalent values as you work with students to model this flexible thinking.

DIFFERENTIATE INSTRUCTION

Consider providing a number line for students to use when working through question 4 to provide a framework for comparing values for those who need it.

1.3.4 GUIDED PRACTICE: ABSOLUTE VALUE

TEACHER MOVES

For questions 1 and 2, have students work independently to answer the questions, stopping after question 2 to discuss student responses. Partners can then work together to complete questions 3 – 5. Bring the whole group together to share and discuss answers.

TEACHER MOVES

Prompt students to connect distance from zero to the absolute value of a number. Monitor for exemplar student responses and elicit those responses from the whole group. For example: “The distance -5 is from 0 is larger than the distance 4 is from 0.”



1.3.4 Guided Practice: Absolute Value

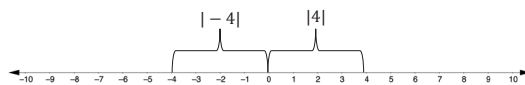
- Using words from the bank below, complete the statements. Not all words will be used.

zero integer distance absolute value

Pairs of opposite values are the same distance from zero on the number line. They are mirror images of each other, with a line of symmetry through zero. The distance a number is from zero on a number line is called absolute value.

- Complete the statement.

The absolute value of 4 and -4 is 4 because they are both 4 units from 0 on the number line.



The symbol for absolute value is two vertical bars around a number. The absolute value of 4 is written as $|4|$.

$$|4| = 4 \text{ and } |-4| = 4$$

- Compare each pair of values and their absolute values.
 - For each row, use the symbols $>$, $<$, or $=$ to complete each inequality.

	Inequality	Absolute Value Inequality
Pair 1	$4 > -5$	$ 4 < -5 $
Pair 2	$-3 < 2$	$ -3 > 2 $
Pair 3	$6 > -6$	$ 6 = -6 $

- b. Discuss what you notice about the comparisons with your partner. Then, summarize your discussion.

Answers may vary. When comparing absolute values, we look at the magnitude or the bigger "number" even though it may be negative since absolute value is looking at the distance from zero.

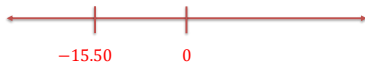
4. Complete the statement.

When comparing the absolute values of two numbers, numbers that are closer to zero will always have the

☐ larger
☒ smaller absolute value.

5. Dariel owes his friend \$15.50.

- a. Sketch a number line that represents this situation.



- b. Represent this situation as a rational number.
 -15.50
- c. What is the absolute value of the rational number?
 $|-15.50| = 15.50$
- d. What does 0 represent in this situation?
Answers will vary. 0 represents when Dariel owes his friend \$0.

1.3.4 GUIDED PRACTICE: ABSOLUTE VALUE (CONTINUED)

TEACHER MOVES

Observe students as they answer question 5b. Students may only consider improper fractions to be acceptable because the definition of a rational number is “numbers that can be expressed as the ratio of two integers.” This is an opportunity to encourage student understanding of the equivalence of numeric values while helping them understand that the definition states that it “can be” expressed as the ratio of two integers, not that it has to be.

QUESTIONING

For a challenge, have students write -15.75 in as many different forms of a rational number as they can think of.

1.3.5 TRY IT: ABSOLUTE VALUE

TEACHER MOVES

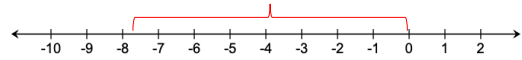
Students may respond to question 2 by noting that two numbers that have the same absolute value are opposites. Prompt them to explain what that means with regard to distance from zero.



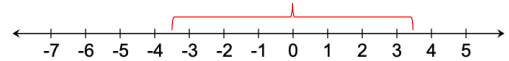
1.3.5 Try It: Absolute Value

1. Complete each statement and model it on the number line given.

a. $|-7.8| = \underline{7.8}$



b. $|\underline{-3\frac{1}{2}} \text{ or } \underline{3\frac{1}{2}}| = 3\frac{1}{2}$



2. What must be true about two numbers that have the same absolute value?
Both numbers must have the same distance away from zero.
3. Use the symbols $>$, $<$, or $=$ to compare each pair of values.

$|-42| \underline{>} -25$ $-21 \underline{<} |18|$ $|-582| \underline{<} |-790|$

**1.3.6 Wrap-Up: Reflection**

1. Complete the learning pyramid.
Answers will vary.

One question I would like to have answered...	
One mistake I learned from today...	One thing I am not sure about...YET!
One thing I am confident I know is...	

1.3.6 WRAP-UP: REFLECTION**DIFFERENTIATE INSTRUCTION**

For additional information, consider prompting students to use Mark-Up text to label responses in the Wrap-Up to corresponding questions or activities. For example, students can label the “confident” box with a check mark and mark any activity or question throughout the lesson that they were confident in with a check mark.